

ICT702 Data Visualisation

Exploring Data Visually

Module 6





I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



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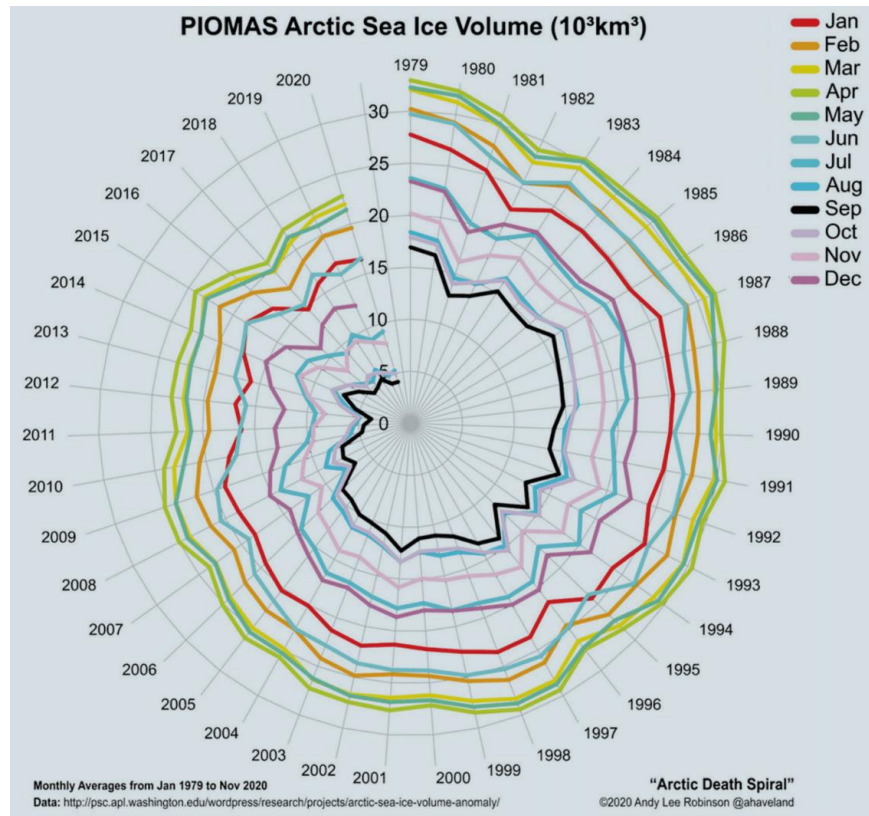
Learning Objectives for Module 6

- Organize and arrange data to facilitate data exploration and visualization.
- Create and interpret visualizations to explore the distributions of individual variables.
- Construct crosstabulations and their related charts to explore patterns in data involving multiple variables.
- Select an appropriate chart type for exploring patterns based on the type of data and the goal of analysis
- Create and compare visualizations that display the association between two quantitative variables.
- Define correlation and how to estimate its strength on a scatter chart.
- Define the types of missing data and understand its implication on how to address it.
- Create visualizations for time series data and identify temporal patterns.
- Explain the strengths and weaknesses of choropleth maps and cartograms for exploring data

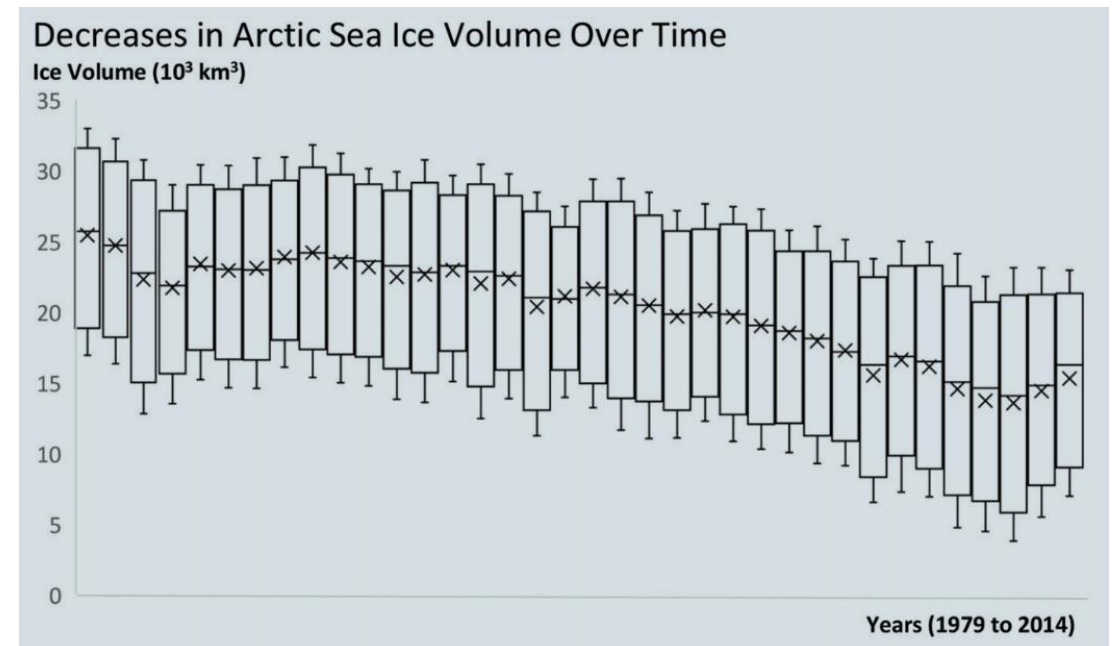
Part 1

Data Visualization Makeover

Radar chart of monthly ice volume observations



Annual box and whisker plots of monthly ice volume observations



Exploratory Data Analysis (EDA)

Understanding data through heavy use of descriptive statistics and visualization

Objectives include:

- detection of errors / missing values / other unusual observations
- characterisation of distribution of the values for the individual variables
- identification of patterns and relationships

Structural dimensions of the data must be considered in EDA:

- **Tall data** = many records (rows) of data
- **Wide data** = many variables (columns)

Espléndido Jugo y Batido, Inc. (EJB) Example

A portion of Espléndido Jugo y Batido, Inc. data in the *EJB* file

	A	B	C	D	E	F	G	H	I	J
1	Order ID	Flavor	Category	\$ Sales	Date Ordered	Date Delivered	Distribution Center	New Customer?	Service Satisfaction Rating	Product Satisfaction Rating
2	92145	Beet	Juices	\$605.97	1/1/2018	1/3/2018	ID	Yes	3	
3	92145	Apple	Juices	\$1,549.00	1/1/2018	1/3/2018	ID	Yes	5	4
4	92145	Apple	Juices	\$1,986.28	1/1/2018	1/3/2018	ID	Yes	5	4
5	92145	Orange	Juices	\$16.43	1/1/2018	1/3/2018	ID	Yes	5	3
6	92145	Cucumber	Juices	\$1,594.93	1/1/2018	1/3/2018	ID	Yes		
7	92145	Pear	Juices	\$590.15	1/1/2018	1/3/2018	ID	Yes		5

- Each data record consists of one flavour's amount included in one order.
- An order can consist of multiple records in the data.

EJB Example: Display Missing Data

EJB data conditionally formatted to display pattern of missing data

	A	B	C	D	E	F	G	H	I	J
1	Order ID	Flavor	Category	\$ Sales	Date Ordered	Date Delivered	Distribution Center	New Customer?	Service Satisfaction Rating	Product Satisfaction Rating
2	92145	Beet	Juices	605.97	1/1/2018	1/3/2018	ID	Yes	3	
3	92145	Apple	Juices	1549.00	1/1/2018	1/3/2018	ID	Yes	5	4
4	92145	Apple	Juices	1986.28	1/1/2018	1/3/2018	ID	Yes	5	4
5	92145	Orange	Juices	16.43	1/1/2018	1/3/2018	ID	Yes	5	3
6	92145	Cucumber	Juices	1594.93	1/1/2018	1/3/2018	ID	Yes		
7	92145	Pear	Juices	590.15	1/1/2018	1/3/2018	ID	Yes		5
8	92146	Beet	Juices	1764.92	1/1/2018	1/3/2018	MS	Yes	3	
9	92146	Pear	Juices	428.46	1/1/2018	1/3/2018	MS	Yes	5	
10	92146	Cucumber	Smoothies	138.92	1/1/2018	1/3/2018	MS	Yes		3
11	92146	Orange	Juices	6.50	1/1/2018	1/3/2018	MS	Yes		4
12	92147	Beet	Juices	872.00	1/1/2018	1/3/2018	ND	No	3	
13	92147	Orange	Smoothies	1437.94	1/1/2018	1/3/2018	ND	No	3	
14	92150	Grape	Juices	1366.97	1/1/2018	1/3/2018	NE	Yes	3	5
15	92151	Apple	Juices	1786.11	1/1/2018	1/3/2018	NE	Yes	5	
16	92152	Orange	Smoothies	63.72	1/1/2018	1/3/2018	NM	Yes	3	3
17	92152	Tomato	Smoothies	1971.52	1/1/2018	1/3/2018	NM	Yes	3	
18	92153	Carrot	Smoothies	1366.25	1/1/2018	1/3/2018	NM	No	5	
19	92737	Orange	Juices	920.78	3/10/2018	3/12/2018	NE	No	3	
20	92974	Orange	Juices	81.29	3/29/2018	3/31/2018	RI	No	4	
21	92156	Orange	Juices	312.87	1/1/2018	1/3/2018	RI	Yes		4
22	93205	Orange	Juices	543.54	4/19/2018	4/21/2018	MS	No	3	
23	92199	Apple	Juices	1128.87	1/5/2018	1/7/2018	MS	No		
24	92199	Orange	Smoothies	1629.45	1/5/2018	1/7/2018	MS	No		
25	92245	Pear	Smoothies	1982.27	1/10/2018	1/12/2018	MS	Yes		
26	92246	Tomato	Smoothies	1624.45	1/10/2018	1/12/2018	ND	No	4	
27	92247	Carrot	Juices	411.28	1/10/2018	1/12/2018	ND	No	4	5
28	92247	Orange	Smoothies	443.48	1/10/2018	1/12/2018	ND	No		
29	92252	Celery	Juices	215.24	1/10/2018	1/12/2018	WV	No	4	
30	92252	Pear	Juices	1431.32	1/10/2018	1/12/2018	WV	No	4	
31	92253	Cucumber	Juices	1212.89	1/10/2018	1/12/2018	WV	No	4	
32	92253	Tomato	Juices	1730.99	1/10/2018	1/12/2018	WV	No	4	
33	92265	Grape	Smoothies	36.91	1/12/2018	1/14/2018	ID	No	5	2
34	92282	Beet	Juices	1602.19	1/13/2018	1/15/2018	WV	No	4	
35	92282	Tomato	Juices	797.25	1/13/2018	1/15/2018	WV	No	4	2
36	93213	Orange	Juices	1013.00	4/19/2018	4/21/2018	NE	No	3	
37	92298	Cucumber	Juices	928.99	1/16/2018	1/18/2018	ND	Yes	1	5
38	92298	Carrot	Smoothies	303.48	1/16/2018	1/18/2018	ND	Yes	1	2



EJB data Formatting as an Excel table

Portion of table showing all records

	A	B	C	D	E	F	G	H	I	J	K
1	Order ID ▼	Flavor ▼	Category ▼	\$ Sales ▼	Date Ordered ▼	Date Delivered ▼	Time To Deliver ▼	Distribution Center ▼	New Customer? ▼	Service Satisfaction Rating ▼	Product Satisfaction Rating ▼
2	92145	Beet	Juices	605.97	1/1/2018	1/3/2018	1/2/1900	ID	Yes	3	
3	92145	Apple	Juices	1549.00	1/1/2018	1/3/2018	1/2/1900	ID	Yes	5	4
4	92145	Apple	Juices	1986.28	1/1/2018	1/3/2018	1/2/1900	ID	Yes	5	4
5	92145	Orange	Juices	16.43	1/1/2018	1/3/2018	1/2/1900	ID	Yes	5	3
6	92145	Cucumber	Juices	1594.93	1/1/2018	1/3/2018	1/2/1900	ID	Yes		

Portion of table filtered to show Beet flavour smoothies

	A	B	C	D	E	F	G	H	I	J
1	Order ID ▼	Flavor ▼	Category ▼	\$ Sales ▼	Date Ordered ▼	Date Delivered ▼	Distribution Center ▼	New Customer? ▼	Service Satisfaction Rating ▼	Product Satisfaction Rating ▼
41	92301	Beet	Smoothies	1180.36	1/16/2018	1/18/2018	NE	No	1	
51	92311	Beet	Smoothies	1441.67	1/16/2018	1/18/2018	WV	No	1	
54	92370	Beet	Smoothies	44.72	1/25/2018	1/27/2018	NE	No	3	4
118	92651	Beet	Smoothies	1315.08	3/2/2018	3/4/2018	NM	No	2	
263	93457	Beet	Smoothies	651.19	5/9/2018	5/11/2018	ID	No	5	4

EJB data Filtering and Sorting Examples

Filtered to show only beet smoothie records with sales $\geq$ \$1,500

	A	B	C	D	E	F	G	H	I	J
1	Order ID ▼	Flavor ▼	Category ▼	\$ Sales ▼	Date Ordered ▼	Date Delivered ▼	Distribution Center ▼	New Customer? ▼	Service Satisfaction Rating ▼	Product Satisfaction Rating ▼
279	93588	Beet	Smoothies	1663.03	5/21/2018	5/23/2018	ND	No	5	
357	94056	Beet	Smoothies	1518.84	7/11/2018	7/13/2018	ND	No	2	
1529	100582	Beet	Smoothies	1740.09	1/10/2020	1/12/2020	ND	No	4	
1921	102392	Beet	Smoothies	1500.04	5/5/2020	5/7/2020	ND	No	5	
3368	94785	Beet	Smoothies	1620.61	9/27/2018	9/30/2018	ID	No	3	

Sorted oldest to newest date delivered

	A	B	C	D	E	F	G	H	I	J
1	Order ID ▼	Flavor ▼	Category ▼	\$ Sales ▼	Date Ordered ▼	Date Delivered ▼	Distribution Center ▼	New Customer? ▼	Service Satisfaction Rating ▼	Product Satisfaction Rating ▼
2	92145	Beet	Juices	605.97	1/1/2018	1/3/2018	ID	Yes	3	
3	92145	Apple	Juices	1549.00	1/1/2018	1/3/2018	ID	Yes	5	4
4	92145	Apple	Juices	1986.28	1/1/2018	1/3/2018	ID	Yes	5	4
5	92145	Orange	Juices	16.43	1/1/2018	1/3/2018	ID	Yes	5	3
6	92145	Cucumber	Juices	1594.93	1/1/2018	1/3/2018	ID	Yes		

EJB data Examples: Adding Variables

Adding a Calculated Field Variable: *Time to Deliver*

	A	B	C	D	E	F	G	H	I	J	K
1	Order ID ▼	Flavor ▼	Category ▼	\$ Sales ▼	Date Ordered ▼	Date Delivered ▼	Time To Deliver ▼	Distribution Center ▼	New Customer? ▼	Service Satisfaction Rating ▼	Product Satisfaction Rating ▼
2	92145	Beet	Juices	605.97	1/1/2018	1/3/2018	2	ID	Yes	3	
3	92145	Apple	Juices	1549.00	1/1/2018	1/3/2018	2	ID	Yes	5	4
4	92145	Apple	Juices	1986.28	1/1/2018	1/3/2018	2	ID	Yes	5	4
5	92145	Orange	Juices	16.43	1/1/2018	1/3/2018	2	ID	Yes	5	3
6	92145	Cucumber	Juices	1594.93	1/1/2018	1/3/2018	2	ID	Yes		

Adding Time Variables: *Year, Month, and Day Ordered*

	A	B	C	D	E	F	G	L	M	N
1	Order ID ▼	Flavor ▼	Category ▼	\$ Sales ▼	Date Ordered ▼	Date Delivered ▼	Time To Deliver ▼	Year Ordered ▼	Month Ordered ▼	Day Ordered ▼
2	92145	Beet	Juices	605.97	1/1/2018	1/3/2018	2	2018	Jan	1
3	92145	Apple	Juices	1549.00	1/1/2018	1/3/2018	2	2018	Jan	1
4	92145	Apple	Juices	1986.28	1/1/2018	1/3/2018	2	2018	Jan	1
5	92145	Orange	Juices	16.43	1/1/2018	1/3/2018	2	2018	Jan	1
6	92145	Cucumber	Juices	1594.93	1/1/2018	1/3/2018	2	2018	Jan	1

EJB data Examples: Average Values

Average \$ Sales for all records using table's Total Row

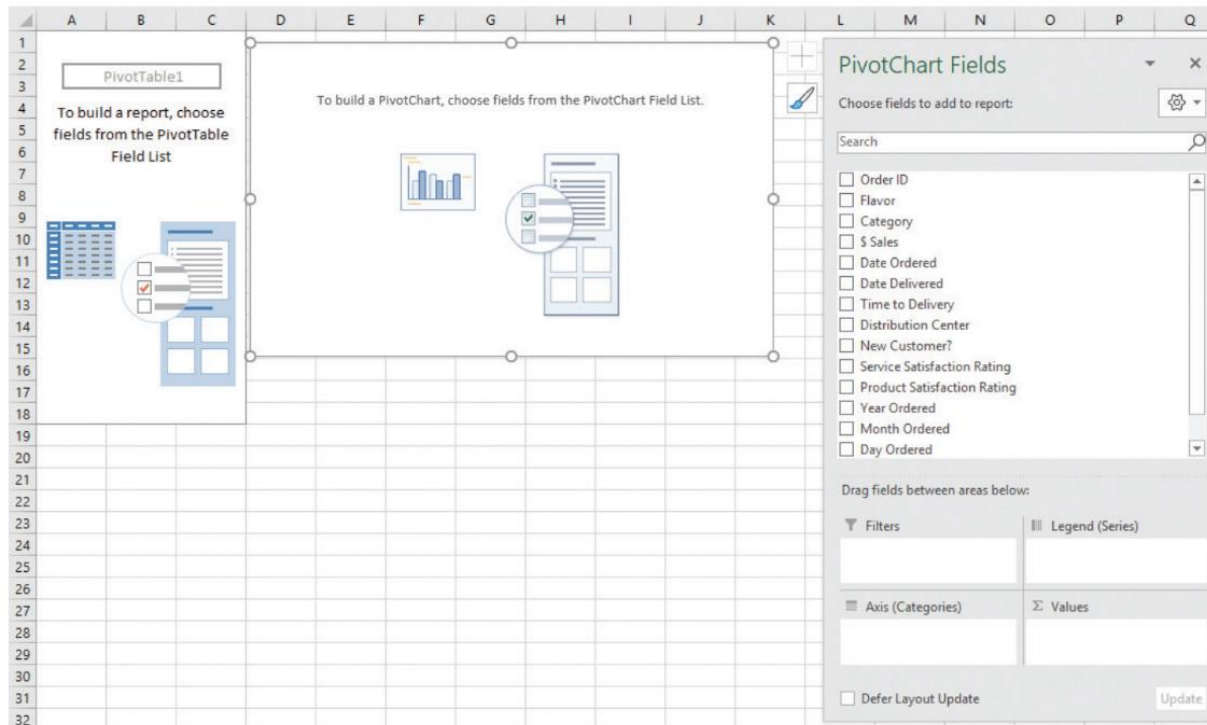
	Order ID	Flavor	Category	\$ Sales	Date Ordered	Date Delivered	Time To Deliver	Distribution Center	New Customer?
19926	105012	Tomato	Juices	1889.96	10/16/2020	11/6/2020	21	ND	Yes
19927	105012	Beet	Smoothies	1111.82	10/16/2020	11/6/2020	21	ND	Yes
19928	105012	Cucumber	Juices	1576.28	10/16/2020	11/6/2020	21	ND	Yes
19929	105012	Apple	Juices	1743.86	10/16/2020	11/6/2020	21	ND	Yes
19930	105017	Beet	Juices	90.05	10/16/2020	11/6/2020	21	NE	No
19931	105017	Grape	Juices	1375.50	10/16/2020	11/6/2020	21	NE	No
19932	105029	Pear	Juices	1486.91	10/16/2020	11/6/2020	21	WV	No
19933	Total			894.18					

Average \$ Sales for beet smoothie records using table's Total Row

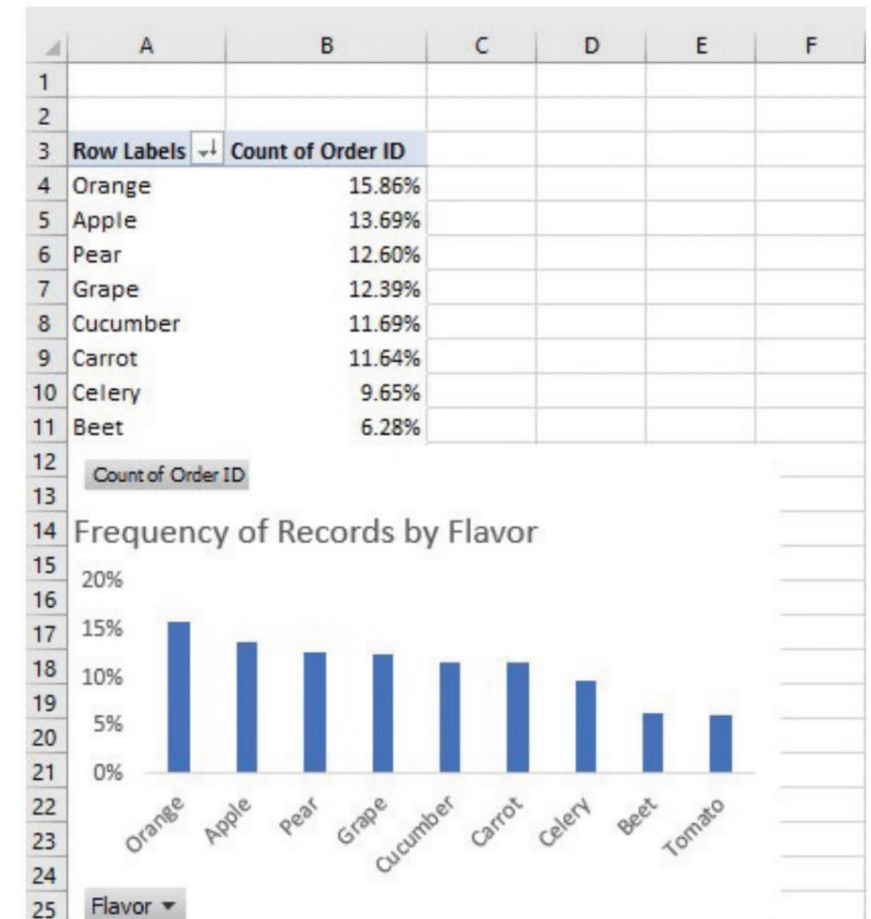
	Order ID	Flavor	Category	\$ Sales	Date Ordered	Date Delivered	Time To Deliver	Distribution Center	New Customer?
19791	104488	Beet	Smoothies	130.13	9/15/2020	9/26/2020	11	NM	No
19805	103559	Beet	Smoothies	1225.41	7/20/2020	8/4/2020	15	NE	No
19861	94183	Beet	Smoothies	196.77	7/24/2018	8/10/2018	17	MS	Yes
19873	95838	Beet	Smoothies	362.66	1/9/2019	1/26/2019	17	ID	Yes
19923	106068	Beet	Smoothies	939.72	12/27/2020	1/14/2021	18	NM	No
19927	105012	Beet	Smoothies	1111.82	10/16/2020	11/6/2020	21	ND	Yes
19933	Total			706.05					

Exploring a Categorical Variable: Pivot Tables / Pivot Charts

EJB data: PivotTable and PivotChart



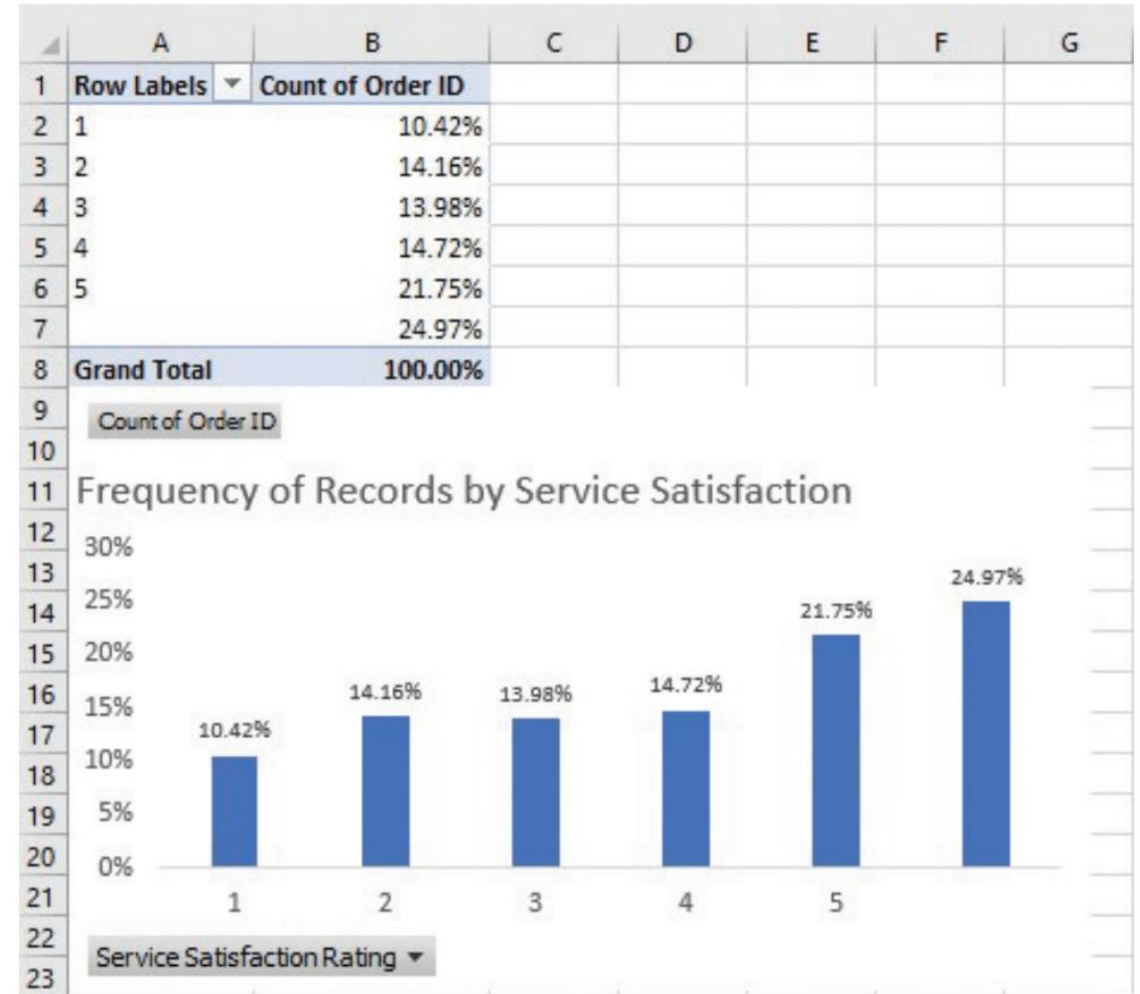
Flavour distribution



EJB data: Exploring an Ordinal Categorical Variable

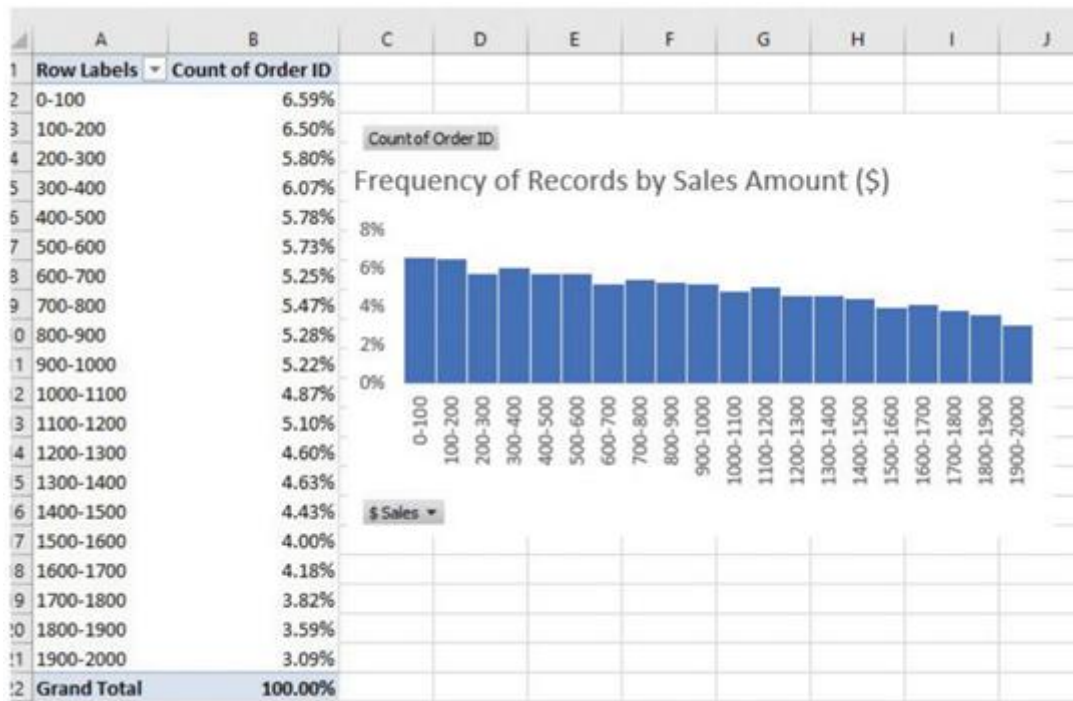
PivotTable and PivotChart for *Service Satisfaction Rating*

When building Frequency charts for ordinal variables, do not disrupt the natural order of the variable values.

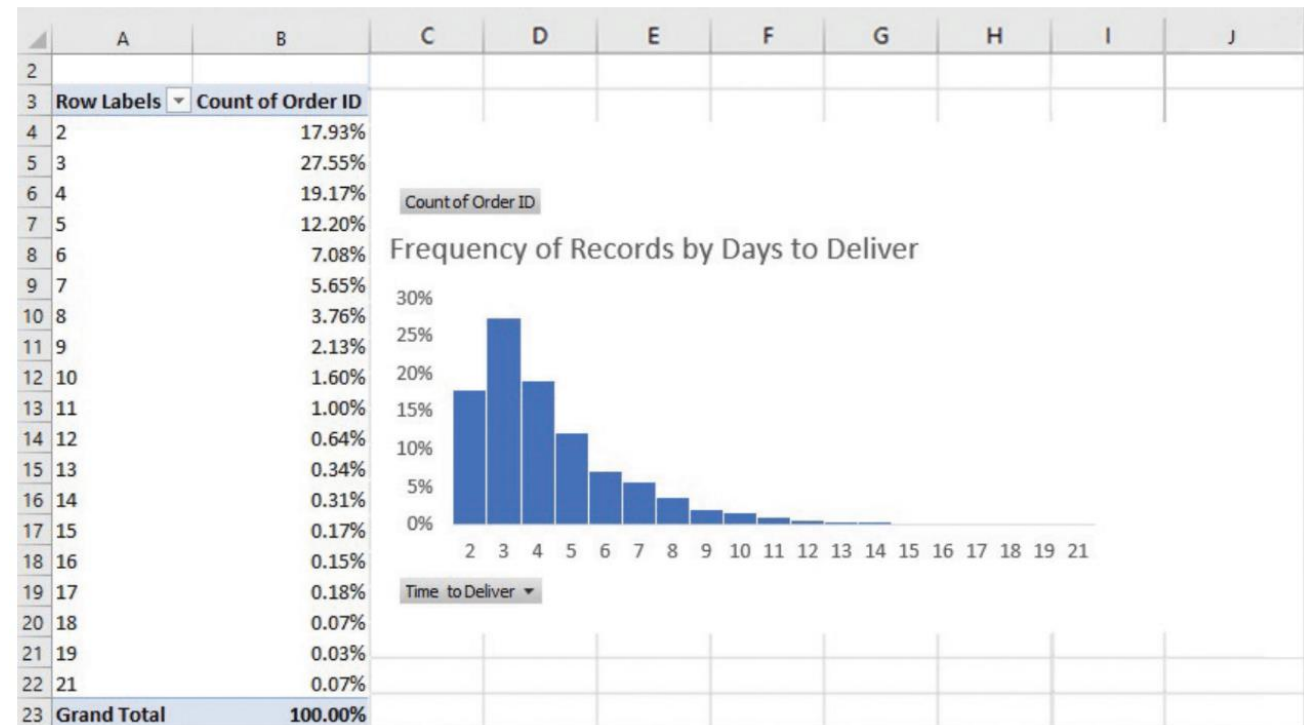


EJB data: Exploring a Quantitative Variable

Flavour distribution of *\$ Sales* at product record level

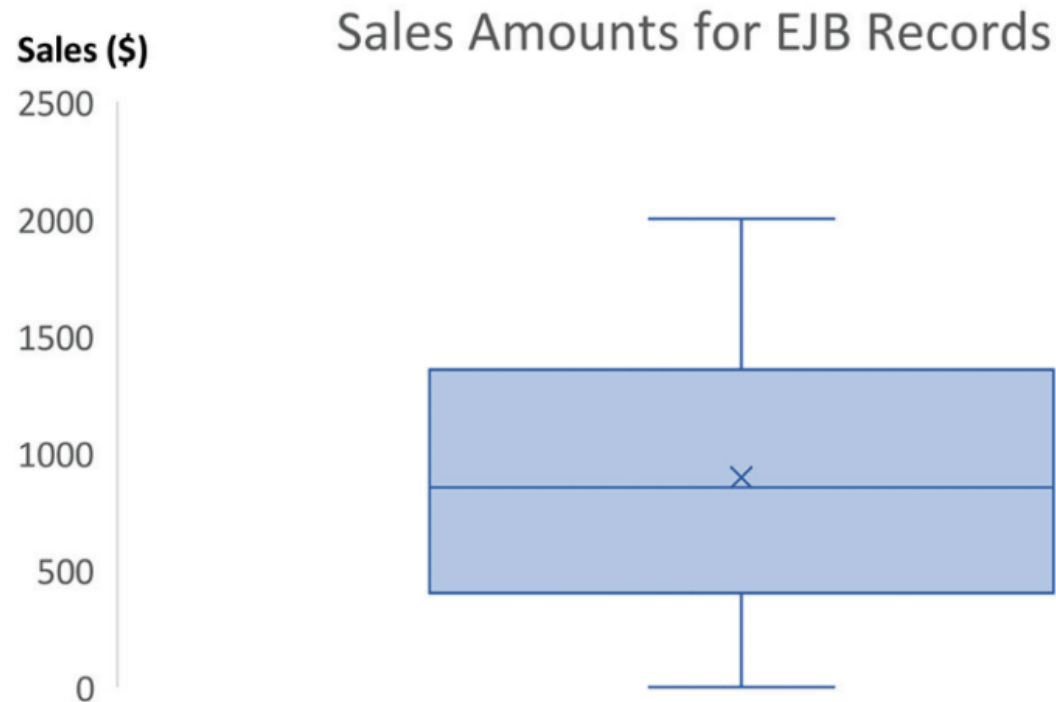


Distribution of *Time to Deliver* at the product record level



Exploring a Quantitative Variable: Box and Whisker Plot

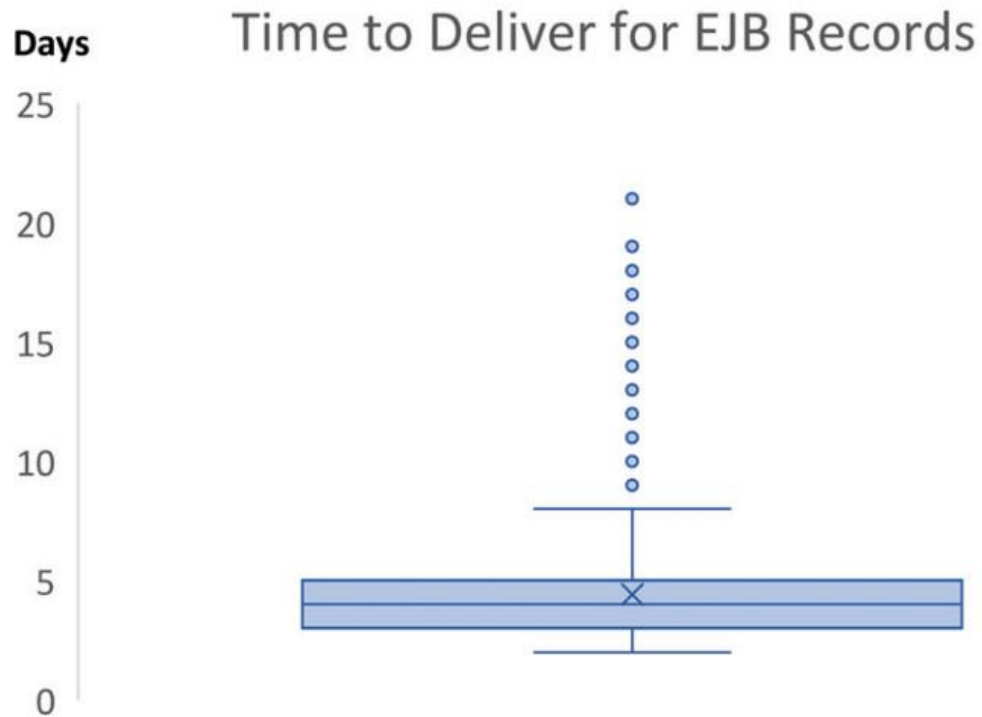
Box and whisker plot of \$ Sales at the product record level in the *EJB* data



Upper whisker is a bit longer than lower whisker therefore distribution of \$ Sales has a slight positive skew.

Box and Whisker Plot for a Skewed Quantitative Variable

Box and whisker plot of *Time to Deliver* at the product record level in the *EJB* data



Distribution of **Time to Deliver** has a pronounced positive skew as the upper whisker is considerably longer than the lower whisker.

End of Part 1